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Drain Valve Assembly

Abstract

A drain valve assembly contains a body including a chamber, at least one rotatable stop portion, a first receiving orifice, a discharge orifice, a coupling orifice, and a second receiving orifice. A shaftless ball valve is received in the chamber and includes a fixing orifice, an opening, a first draining orifice, a second draining orifice, a third draining orifice and a stopping face. A first reinforcement ring is mounted on the upper support portion and the lower support portion. A second reinforcement ring is located at an upper portion of the chamber. A positioning ring includes a through orifice and an abutting portion. A controller includes a connection portion and an operation portion. A limiter is configured to connect the controller with the coupling orifice, and a filtration holder disposed on the first receiving orifice.

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to a drain valve assembly which is fixed below a water sink to obtain a shafting of stopping and draining waters.

BACKGROUND

[0002] A conventional drain valve assembly is disclosed in U.S. Pat. No. 6,058,526, but such a drain valve assembly is manufactured difficultly. As illustrated in FIGS. 4 and 7 of U.S. Pat. No. 6,058,526, the conventional drain valve assembly contains a hollow ball valve, and the hollow ball valve includes a socket fixed on an end thereof, a roller connected on the other end of the hollow ball valve, and a drainage hole and a flat surface which are defined on the hollow ball valve, wherein the flat surface is beveled from a quarter spherical shape. The hollow ball valve is accommodated in a body and is machined for the discharge and the beveled surface process. Also, the hollow ball valve includes the roller connected with a trunion. A retainer includes a retainer bar received in an opening thereof and configured to rotatably operate the retainer easily. However, it is easy to block impurities in the retainer to reduce water discharge from the conventional drain valve assembly. In addition, the body of the hollow ball valve is machined and connected troublesomely to increase the machining cost and connecting difficulty.

[0003] The present invention has arisen to mitigate and/or obviate the afore-described disadvantages.

SUMMARY

[0004] The primary aspect of the present invention is to provide a drain valve assembly by which the discharge orifice has the lower support portion extending to the chamber from a lower end thereof so that the first reinforcement ring and the second reinforcement ring are mounted on the shaftless ball valve in the chamber, thus machining and connecting the body easily.

[0005] Another aspect of the present invention is to provide a drain valve assembly by which the shaftless ball valve includes the defining orifice configured to match with the limiting post, thus machining the shaftless ball valve quickly to reduce a defective rate greatly.

[0006] To obtain the above-mentioned aspects, a drain valve assembly provided by the present invention contains a body including a chamber formed in the body and having at least one rotatable stop portion, a first receiving orifice defined on a top of the body and communicating with the chamber, a discharge orifice formed on a side of the body, a coupling orifice defined on the other side of the body and communicating with the chamber, and a second receiving orifice formed on a bottom of the body and communicating with the chamber.

[0007] The first receiving orifice has a first shoulder formed on an outer wall of the first receiving orifice, and a second shoulder and a first threaded section which are formed on an inner wall of the first receiving orifice.

[0008] The discharge orifice has a lower support portion extending to the chamber from a lower end thereof so that a profile of the discharge orifice is downward changed from a circle shape to a non-circle shape to connect with the chamber by using an upper end of the discharge orifice, thus supplying waters.

[0009] The second receiving orifice includes an upper support portion formed on a top thereof and located in the chamber and on the lower support portion.

[0010] A shaftless ball valve is received in the chamber and includes a fixing orifice, an opening corresponding to the fixing orifice, a first draining orifice, a second draining orifice, a third draining orifice and a stopping face which are formed in turn between the fixing orifice and the opening. The opening, the first draining orifice, the second draining orifice, and the third draining orifice are communicated with one another. The shaftless ball valve further includes a defining orifice corresponding to the rotatable stop portion, a limiting post configured to match with the defining orifice. The limiting post has a locating section formed on an end thereof and a defending section formed on the other end of the limiting post, the location section is connected with the

defining orifice, and the defending section extrudes out of the shaftless ball valve.

[0011] A first reinforcement ring is mounted on the upper support portion and the lower support portion, and the first reinforcement ring is located below the shaftless ball valve.

[0012] A second reinforcement ring is located at an upper portion of the chamber and above the shaftless ball valve.

[0013] A positioning ring includes a through orifice defined on a circular center of the positioning ring, a second threaded section formed on an outer wall of the positioning ring and configured to screw with the first threaded section, and an abutting portion is fixed on a bottom of a limiter.

[0014] A controller includes a connection portion formed on an end of the controller and connected with the fixing orifice of the shaftless ball valve.

[0015] A limiter is configured to connect the controller with the coupling orifice.

[0016] A filtration holder is disposed on the first receiving orifice.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] FIG. 1 is a perspective view showing the assembly of a drain valve assembly according to a preferred embodiment of the present invention.

[0018] FIG. 2 is a perspective view showing the exploded components of the drain valve assembly according to the preferred embodiment of the present invention.

[0019] FIG. 3 is another perspective view showing the exploded components of the drain valve assembly according to the preferred embodiment of the present invention.

[0020] FIG. 4 is another perspective view showing the assembly of the drain valve assembly according to the preferred embodiment of the present invention.

[0021] FIG. 5 is a top plan view showing the assembly of the drain valve assembly according to the preferred embodiment of the present invention.

[0022] FIG. 6 is a cross sectional view taken along the line A-A of FIG. 5.

[0023] FIG. 7 is a cross sectional view showing the operation of the drain valve assembly according to the preferred embodiment of the present invention.

[0024] FIG. 8 is another cross sectional view showing the operation of the drain valve assembly according to the preferred embodiment of the present invention.

[0025] FIG. 9 is also another cross sectional view showing the operation of the drain valve assembly according to the preferred embodiment of the present invention.

[0026] FIG. 10 is a cross sectional view taken along the line B-B of FIG. 5.

[0027] FIG. 11 is still another cross sectional view showing the operation of the drain valve assembly according to the preferred embodiment of the present invention.

[0028] FIG. 12 is a perspective view showing the operation of a filtration holder of the drain valve assembly according to the preferred embodiment of the present invention.

[0029] FIG. 13 is another perspective view showing the operation of the filtration holder of the drain valve assembly according to the preferred embodiment of the present invention.

[0030] FIG. 14 is also another perspective view showing the operation of the filtration holder of the drain valve assembly according to the preferred embodiment of the present invention.

[0031] FIG. 15 is a cross sectional view showing the operation of the filtration holder of the drain valve assembly and taken along the line C-C of FIG. 5.

DETAILED DESCRIPTION

[0032] With reference to FIGS. 1-6, a drain valve assembly according to a preferred embodiment of the present invention comprises: a body 10, a chamber 11 formed in the body 10 and having at least one rotatable stop portion 111, a first receiving orifice 12 defined on a top of the body 10 and communicating with the chamber 11, wherein a diameter of the first receiving orifice 12 is equal to

or greater than a diameter of the chamber **11**, a discharge orifice **13** formed on a side of the body **10**, a coupling orifice **14** defined on the other side of the body **10** and communicating with the chamber **11**, and a second receiving orifice **15** formed on a bottom of the body **10** and communicating with the chamber **11**.

[0033] The first receiving orifice **12** has a first shoulder **121** formed on an outer wall thereof, and a second shoulder **122** and a first threaded section **123** which are formed on an inner wall of the first receiving orifice **12**.

[0034] The discharge orifice **13** has a lower support portion **131** extending to the chamber **11** from a lower end thereof so that a profile of the discharge orifice **13** is downward changed from a circle shape to a non-circle shape to connect with the chamber **11** by using an upper end of the discharge orifice **13**, thus supplying waters.

[0035] The second receiving orifice **15** includes an upper support portion **151** formed on a top thereof and located in the chamber **11** and on the lower support portion **131**.

[0036] A shaftless ball valve **20** is received in the chamber **11** and includes a fixing orifice **21**, an opening **22** corresponding to the fixing orifice **21**, a first draining orifice **23**, a second draining orifice **24**, a third draining orifice **25** and a stopping face **26** which are formed in turn between the fixing orifice **21** and the opening **22**, wherein the opening **22**, the first draining orifice **23**, the second draining orifice **24**, and the third draining orifice **25** are communicated with one another. The shaftless ball valve **20** further includes a defining orifice **27** corresponding to the rotatable stop portion **111**, a limiting post **28** configured to match with the defining orifice **27**, wherein the limiting post **28** has a locating section **281** formed on an end thereof and a defending section **282** formed on the other end of the limiting post **28**, wherein the location section **281** is in a threaded column shape and is screwed with the defining orifice **27**, and the defending section **282** is a square column and extrudes out of the shaftless ball valve **20**.

[0037] A first reinforcement ring **30** is annular and is mounted on the upper support portion **151** and the lower support portion **131**, wherein the first reinforcement ring **30** includes an arcuate face defined on a top thereof and located below the shaftless ball valve **20**, and the first reinforcement ring **30** includes a reinforced loop **31** connected on a bottom thereof.

[0038] A second reinforcement ring **40** is annular and includes an arcuate face defined on a bottom thereof, and the second reinforcement ring **40** includes a first seal loop **41** connected on an outer wall thereof and is located at an upper portion of the chamber **11** and above the shaftless ball valve **20**, wherein the second reinforcement ring **40** includes a groove **42** defined on a top thereof and a spring **43** received in the groove **42**.

[0039] A positioning ring **50** includes a through orifice **51** defined on a circular center thereof, a second threaded section **52** formed on an outer wall of the positioning ring **50** and configured to screw with the first threaded section **123**, and an abutting portion **53** fixed on a bottom of a limiter **70**, wherein a tilted face extends downward to a center circular center of the positioning ring **50** and between the through orifice **51** and the outer wall of the positioning ring **50**, wherein the tilted face has at least one projection **511**, at least one first engagement foot **54** extending from the projection **511**, at least one locking orifice **55** locked with a screw **56**, and at least one shrink portion **57** located between the at least one first engagement foot **54** and the locking orifice **55**, wherein the at least one first engagement foot **54**, the at least one locking orifice **55** and the at least one shrink portion **57** are concentric with the through orifice **51**.

[0040] A controller **60** includes a connection portion **61** formed on an end thereof, an operation portion **62** formed on the other end of the controller **60**, and a retainer **63** disposed adjacent to the connection portion **61**, wherein the connection portion **61** is connected with the fixing orifice **21** of the shaftless ball valve **20** so that the controller **60** controls the shaftless ball valve **20** to rotate.

[0041] The limiter **70** includes a second seal loop **71**, and the limiter **70** is configured to connect the retainer **63** of the controller **60** with the coupling orifice **14** and to stop the waters by using the second seal loop **71**.

[0042] A filtration holder **80** is disposed on the first receiving orifice **12**, wherein the filtration holder **80** includes at least one second engagement foot **81** corresponding to the at least one first engagement foot **54**, and the filtration holder **80** includes at least one passing orifice **82** corresponding to the at least one locking orifice **55**.

[0043] In assembly, the first reinforcement ring **30** is mounted on the upper support portion **151** and the lower support portion **131** by way of the reinforced loop **31**, wherein the first reinforcement ring **30** and the reinforced loop **31** pass through a middle section of the discharge orifice **13**, and the shaftless ball valve **20** is received in the chamber **11** and is engaged with the first reinforcement ring **30**. The controller **60** is coupled with the fixing orifice **21** of the shaftless ball valve **20** by using the connection portion **61** and is received in the coupling orifice **14** by engaging the second seal loop **71** with the retainer **63**. The second reinforcement ring **40** is located above the shaftless ball valve **20** and seals the chamber **11** by way of the first seal loop **41**, and the second threaded section **52** of the positioning ring **50** is screwed with the first threaded section **123**, wherein the positioning ring **50** forces the second reinforcement ring **40** to abut against the shaftless ball valve **20** so as to fix the shaftless ball valve **20**. In the meantime, the defending section **282** is located on the rotatable stop portion **111**. When the operation portion **62** of the controller **60** is operated, the shaftless ball valve **20** is located at a first position X1 (as shown in FIGS. 7-9) or the second position X2 (as shown in FIGS. 6, 10 and 11) to rotate, wherein the filtration holder **80** is mounted in the first receiving orifice **12**, the at least one second engagement foot **81** is aligned with the shrink portion **57** to rotate the filtration holder **80**, such that the at least one second engagement foot **81** is engaged with the at least one first engagement foot **54** so that the locking orifice **55** aligns with the at least one passing orifice **82**, and the screw **56** of the locking orifice **55** is locked with the locking orifice **55** or the screw **56** is locked with the locking orifice **55** via the at least one passing orifice **82** (as illustrated in FIGS. 12-15).

[0044] When turning on the drain valve assembly to drain the waters, as shown in FIGS. 6, 10 and 11, the operation portion **62** is operated to control the shaftless ball valve **20** to rotate. Since the shaftless ball valve **20** is completely covered by the first reinforcement ring **30** and the second reinforcement ring **40**, the shaftless ball valve **20** merely rotates between the first position X1 and the second position X2 of the rotatable stop portion **111** of the chamber **11**, such that the limiting post **28** is rotated to the second position X2 so that the first draining orifice **23** aligns with the first receiving orifice **12**, the second draining orifice **24** aligns with the second receiving orifice **15**, and the discharge orifice **13** aligns with the opening **22**, thus draining the waters.

[0045] Referring to FIGS. 7-9, when turning off the drain valve assembly to stop the waters, the operation portion **62** is operated to control the shaftless ball valve **20** to rotate. Due to the shaftless ball valve **20** is completely covered by the first reinforcement ring **30** and the second reinforcement ring **40**, the shaftless ball valve **20** merely rotates between the first position X1 and the second position X2 of the rotatable stop portion **111** of the chamber **11**, such that the limiting post **28** is rotated to the second position X1 so that the stopping face **26** aligns with the first receiving orifice **12**, the second draining orifice **24** aligns with the second receiving orifice **15**, and the discharge orifice **13** aligns with the opening **22**, thus stopping the waters.

[0046] Accordingly, the drain valve assembly has the following advantages:

[0047] The shaftless ball valve **20** includes multiple orifices, such as the opening **22**, the first draining orifice **23**, the second draining orifice **24**, and the third draining orifice **25** to discharge and overflow the waters. In the meantime, the chamber **11** corresponds to the first receiving orifice **12**, the discharge orifice **13**, the coupling orifice **14**, and the second receiving orifice **15** so that when the shaftless ball valve **20** is received in the chamber **11**, an axial line passes through the chamber **11** and the shaftless ball valve **20**. The discharge orifice **13** has the lower support portion **131** extending to the chamber **11** from the lower end thereof so that the first reinforcement ring **30** and the second reinforcement ring **40** are mounted on the shaftless ball valve **20** in the chamber **11**, thus machining and connecting the body **10** easily. Also, the shaftless ball valve **20** includes the defining

orifice 27 which matches with the limiting post 28, thus machining the shaftless ball valve 20 quickly to reduce a defective rate greatly.

[0048] While the first embodiments of the invention have been set forth for the purpose of disclosure, modifications of the disclosed embodiments of the invention as well as other embodiments thereof may occur to those skilled in the art. The scope of the claims should not be limited by the first embodiments set forth in the examples, but should be given the broadest interpretation consistent with the description as a whole.

Claims

1. A drain valve assembly comprising: a body including a chamber formed in the body and having at least one rotatable stop portion, a first receiving orifice defined on a top of the body and communicating with the chamber, wherein a diameter of the first receiving orifice is equal to or greater than a diameter of the chamber, a discharge orifice formed on a side of the body, a coupling orifice defined on the other side of the body and communicating with the chamber, and a second receiving orifice formed on a bottom of the body and communicating with the chamber; the first receiving orifice having a first shoulder formed on an outer wall of the first receiving orifice, and a second shoulder and a first threaded section which are formed on an inner wall of the first receiving orifice; the discharge orifice has a lower support portion extending to the chamber from a lower end thereof so that a profile of the discharge orifice is downward changed from a circle shape to a non-circle shape to connect with the chamber by using an upper end of the discharge orifice, thus supplying water; the second receiving orifice including an upper support portion formed on a top thereof and located in the chamber and on the lower support portion; a shaftless ball valve received in the chamber and including a fixing orifice, an opening corresponding to the fixing orifice, a first draining orifice, a second draining orifice, a third draining orifice and a stopping face which are formed in turn between the fixing orifice and the opening, wherein the opening, the first draining orifice, the second draining orifice, and the third draining orifice are communicated with one another; the shaftless ball valve further including a defining orifice corresponding to the rotatable stop portion, a limiting post configured to match with the defining orifice, wherein the limiting post has a locating section formed on an end thereof and a defending section formed on the other end of the limiting post, wherein the locating section is connected with the defining orifice, and the defending section extrudes out of the shaftless ball valve; a first reinforcement ring being annular and mounted on the upper support portion and the lower support portion, wherein the first reinforcement ring includes an arcuate face defined on a top thereof and located below the shaftless ball valve; a second reinforcement ring being annular and including an arcuate face defined on a bottom of the second reinforcement ring, and the second reinforcement ring being located at an upper portion of the chamber and above the shaftless ball valve; a positioning ring including a through orifice defined on a circular center of the positioning ring, a second threaded section formed on an outer wall of the positioning ring and configured to screw with the first threaded section, and an abutting portion fixed on a bottom of a limiter, and wherein the positioning ring includes at least one first engagement foot and at least one shrink portion; a controller including a connection portion formed on an end of the controller, an operation portion formed on the other end of the controller, wherein the connection portion is connected with the fixing orifice of the shaftless ball valve so that the controller controls the shaftless ball valve to rotate; the limiter configured to connect the controller with the coupling orifice; a filtration holder disposed on the first receiving orifice, wherein the filtration holder includes at least one second engagement foot; wherein in assembly, the first reinforcement ring is mounted on the upper support portion and the lower support portion, and the shaftless ball valve is received in the chamber and is engaged with the first reinforcement ring, the controller is coupled with the fixing orifice of the shaftless ball valve by using the connection portion and is received in the coupling orifice; the second reinforcement ring

is located above the shaftless ball valve, and the second threaded section of the positioning ring is screwed with the first threaded section, wherein the positioning ring forces the second reinforcement ring to abut against the shaftless ball valve so as to fix the shaftless ball valve; in the meantime, the defending section is located on the rotatable stop portion, when the operation portion of the controller is operated, the shaftless ball valve is located between a first position and a second position to rotate, wherein the filtration holder is mounted on the first receiving orifice, the at least one second engagement foot is aligned with the at least one shrink portion to rotate the filtration holder, such that the at least one second engagement foot is engaged with the at least one first engagement foot.

2. The drain valve assembly as claimed in claim 1, wherein the locating section is in a threaded column shape and is screwed with the defining orifice, and the defending section is a square column.

3. The drain valve assembly as claimed in claim 1, wherein the second reinforcement ring includes a first seal loop connected on an outer wall thereof, and the limiter includes a second seal loop configured to stop the waters between the limiter and the controller.

4. The drain valve assembly as claimed in claim 1, wherein the second reinforcement ring includes a groove defined on a top thereof and a spring received in the groove; the controller includes a retainer disposed adjacent to the connection portion, wherein the limiter is configured to connect the retainer of the controller with the coupling orifice.

5. The drain valve assembly as claimed in claim 1, wherein a tilted face extends downward to a circular center of the positioning ring and between the through orifice and the outer wall of the positioning ring, wherein the tilted face has at least one projection, the at least one first engagement foot extending from the projection, at least one locking orifice locked with a screw, and the at least one shrink portion located between the at least one first engagement foot and the locking orifice, wherein the at least one first engagement foot, the at least one locking orifice and the at least one shrink portion are concentric with the through orifice; the filtration holder includes the at least one second engagement foot corresponding to the at least one first engagement foot, and the filtration holder includes at least one passing orifice corresponding to the at least one locking orifice; when the at least one second engagement foot is engaged with the at least one first engagement foot, the locking orifice aligns with the at least one passing orifice, and the screw of the locking orifice is locked with the locking orifice or the screw is locked with the locking orifice via the at least one passing orifice.

6. A drain valve assembly comprising: a body including a chamber formed in the body and having at least one rotatable stop portion, a first receiving orifice defined on a top of the body and communicating with the chamber, wherein a diameter of the first receiving orifice is equal to or greater than a diameter of the chamber, a discharge orifice formed on a side of the body, a coupling orifice defined on the other side of the body and communicating with the chamber, and a second receiving orifice formed on a bottom of the body and communicating with the chamber; the first receiving orifice having a first shoulder formed on an outer wall of the first receiving orifice, and a second shoulder and a first threaded section which are formed on an inner wall of the first receiving orifice; the discharge orifice has a lower support portion extending to the chamber from a lower end thereof so that a profile of the discharge orifice is downward changed from a circle shape to a non-circle shape to connect with the chamber by using an upper end of the discharge orifice, thus supplying water; the second receiving orifice including an upper support portion formed on a top thereof and located in the chamber and on the lower support portion; a shaftless ball valve received in the chamber and including a fixing orifice, an opening corresponding to the fixing orifice, a first draining orifice, a second draining orifice, a third draining orifice and a stopping face which are formed in turn between the fixing orifice and the opening, wherein the opening, the first draining orifice, the second draining orifice, and the third draining orifice are communicated with one another; the shaftless ball valve further including a defining orifice corresponding to the rotatable

stop portion, a limiting post configured to match with the defining orifice, wherein the limiting post has a locating section formed on an end thereof and a defending section formed on the other end of the limiting post, wherein the locating section is connected with the defining orifice, and the defending section extrudes out of the shaftless ball valve; a first reinforcement ring being annular and mounted on the upper support portion and below the shaftless ball valve, wherein the first reinforcement ring includes a reinforced loop connected on a bottom thereof, and the first reinforcement ring and the reinforced loop pass through a middle section of the discharge orifice; a second reinforcement ring being annular and including an arcuate face defined on a bottom of the second reinforcement ring, and the second reinforcement ring being located at an upper portion of the chamber and above the shaftless ball valve; a positioning ring including a through orifice defined on a circular center of the positioning ring, a second threaded section formed on an outer wall of the positioning ring and configured to screw with the first threaded section, and an abutting portion fixed on a bottom of a limiter, and wherein the positioning ring includes at least one first engagement foot and at least one shrink portion; a controller includes a connection portion and an operation portion formed on the other end of the controller, wherein the connection portion is connected with the fixing orifice of the shaftless ball valve so that the controller controls the shaftless ball valve to rotate; the limiter configured to connect the controller with the coupling orifice; a filtration holder disposed on the first receiving orifice, wherein the filtration holder includes at least one second engagement foot; wherein in assembly, the first reinforcement ring and the reinforced loop are fixed on the upper support portion, and the shaftless ball valve is received in the chamber and is engaged with the first reinforcement ring, the controller is coupled with the fixing orifice of the shaftless ball valve by using the connection portion and is received in the coupling orifice; the second reinforcement ring is located above the shaftless ball valve, and the second threaded section of the positioning ring is screwed with the first threaded section, wherein the positioning ring forces the second reinforcement ring to abut against the shaftless ball valve so as to fix the shaftless ball valve; in the meantime, the defending section is located on the rotatable stop portion, when the operation portion of the controller is operated, the shaftless ball valve is located between a first position and a second position to rotate, wherein the filtration holder is mounted on the first receiving orifice, the at least one second engagement foot is aligned with the at least one shrink portion to rotate the filtration holder, such that the at least one second engagement foot is engaged with the at least one first engagement foot.

7. The drain valve assembly as claimed in claim 6, wherein the locating section is in a threaded column shape and is screwed with the defining orifice, and the defending section is a square column.

8. The drain valve assembly as claimed in claim 6, wherein a tilted face extends downward to a circular center of the positioning ring and between the through orifice and the outer wall of the positioning ring, wherein the tilted face has at least one projection, the at least one first engagement foot extending from the projection, at least one locking orifice locked with a screw, and the at least one shrink portion located between the at least one first engagement foot and the locking orifice, wherein the at least one first engagement foot, the at least one locking orifice and the at least one shrink portion are concentric with the through orifice; the filtration holder includes the at least one second engagement foot corresponding to the at least one first engagement foot, and the filtration holder includes at least one passing orifice corresponding to the at least one locking orifice; when the at least one second engagement foot is engaged with the at least one first engagement foot, the locking orifice aligns with the at least one passing orifice, and the screw of the locking orifice is locked with the locking orifice or the screw is locked with the locking orifice via the at least one passing orifice.
